

Analyzing Racial Bias in COMPAS

Introduction to Machine Learning
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Abstract—This paper investigates racial bias in the COMPAS tool by replicating its risk scores and predicting actual recidivism under race-aware and race-blind conditions. In race-aware configurations, African-American race carries five times more feature importance than Caucasian race. Removing explicit race features does not eliminate disparity; instead, it flips African-American defendants from being overpredicted to underpredicted. Crucially, a classifier trained on actual recidivism replicates a False Positive Rate for African-Americans over twice that of Caucasians, even when blind. This proves that while algorithms introduce directional distortion, racial bias is fundamentally embedded as structural bias within historical criminal history records.

Index Terms—Racial Bias in Data, Machine Learning, COMPAS, Recidivism

I. INTRODUCTION

In the United States criminal justice system, judges often rely on software to help decide if a defendant should be granted bail or kept in jail. One of the most widely used tools for this is called COMPAS (Correctional Offender Management Profiling for Alternative Sanctions), which analyzes a defendant’s background to predict how likely they are to commit another crime in the future.

However, letting an algorithm decide someone’s sentence is highly controversial. The algorithm was found out to be heavily biased against African-American defendants. The software was much more likely to incorrectly label black defendants as “high risk”, while mistakenly labeling white defendants as “low risk”, even when their criminal histories were very similar.

This project aims to investigate this problem. Using the original dataset released by ProPublica from Broward County, Florida, we are attempting to recreate the algorithm’s predictions training our own chosen machine learning models.

Our goal is to understand how these recidivism scores are generated. More importantly, we aim to analyze exactly how and why racial bias exists in these mathematical models and explore whether it is possible to adjust the data or the algorithms to make the predictions fairer without entirely sacrificing their accuracy.

II. STATE OF THE ART

The discussion around algorithmic fairness in the criminal justice system exploded in 2016 following a groundbreaking investigation by ProPublica titled “Machine Bias” [1]. By analyzing the COMPAS scores of thousands of defendants in Broward County, the journalists discovered a significant disparity in the types of mistakes the algorithm was making.

Specifically, ProPublica looked at the False Positive Rate situations where the algorithm flags a defendant as “high risk”, but the person does not go on to commit another crime. They found that this “false alarm” rate was nearly twice as high for African-American defendants compared to Caucasian defendants. Conversely, Caucasian defendants were more likely to be falsely labeled as “low risk” even if they later re-offended.

Shortly after this report, Northpointe (the creators of COMPAS) fired back with their own defense [2]. They argued that their algorithm was indeed fair because it was perfectly “calibrated.” This means that if the algorithm gave a score of 7 to a white defendant and a 7 to a black defendant, both individuals actually had the exact same statistical probability of re-offending.

This sparked a massive debate in the machine learning and statistics community. Researchers like Kleinberg et al. [3] and Chouldechova [4] analyzed the math and proved what is now a fundamental rule of algorithmic fairness: it is mathematically impossible to satisfy both ProPublica’s definition of fairness (equal false positive rates) and Northpointe’s definition of fairness (equal calibration) at the same time, as long as the base rates of arrest differ between two groups.

Because of this “impossibility theorem,” modern research has shifted from trying to make a perfect algorithm into figuring out which trade-offs we are willing to accept. Today, the COMPAS dataset serves as the standard benchmark in data science for testing new bias-mitigation techniques—such as removing proxy variables, weighting the training data, or tweaking the final model thresholds to create fairer outcomes across different demographic groups.

III. DATA

A. Data Description

The data set we used is a csv containing criminal history, jail and prison time, demographics and COMPAS risk scores for defendants from Broward County, available in ProPublica's repository.

It has 53 columns in total. The columns all fall in the following categories:

- a) *Identifiers & Personal Info*: Record identifiers and personal attributes such as name, age, sex and race.
- b) *Criminal History*: Number of prior criminal charges, as well as juvenile offense counts.
- c) *Current Offense Info*: Detailed information of the crime that triggered the COMPAS assessment such as the degree and description of the crime, the offense date, and the dates when the defendant entered and exited jail for this arrest.
- d) *Algorithm Scores*: Information regarding the attribution of both the general and violent recidivism scores.
- e) *Recidivism Outcomes*: Binary flags indicating subsequent arrest, and some details about the new arrest.
- f) *Temporal & Survival Analysis*: Used to account for time spent in the community vs. time incarcerated.

It is worth mentioning that for our reimplementation of the COMPAS algorithm, we only used some attributes from the first three categories mentioned above, reducing the amount of columns significantly. We will expound on this later.

B. Data Visualization

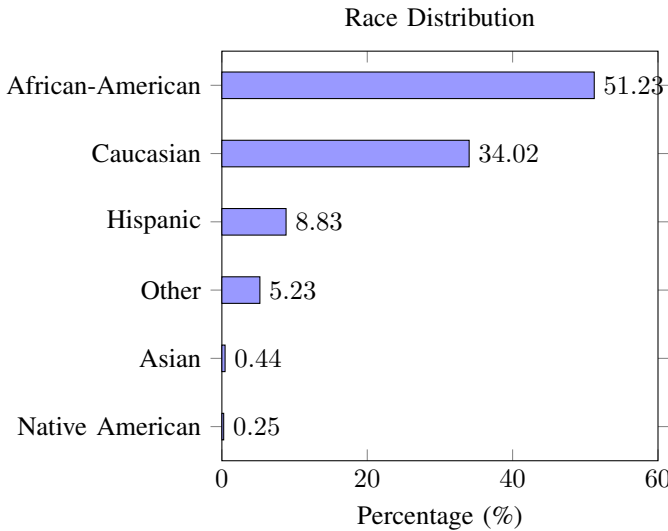


Fig. 1. Race Distribution

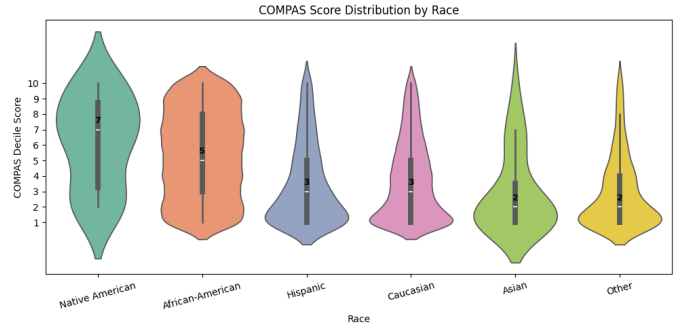


Fig. 2. COMPAS Distribution by Race

As seen in Figure 2, the COMPAS score seems to frequently be higher for defendants of both African-American and Native-American races, being a noticeably lower on the other races. Given the small representation of Native Americans and Asian people, it's hard to draw any meaningful conclusion about the COMPAS score in relation to those races. However, in the case of the African-American and Caucasian races, there's definitely a big disparity.

C. Data Quality

During pre-processing, several data quality issues were identified.

Two pairs of columns were found to be exact duplicates, likely the result of multiple joins across different databases: `decile_score` and `priors_count` each had a duplicate counterpart, and `screening_date` was an exact duplicate of `compas_screening_date`.

Additionally, two columns (`type_of_assessment` and `v_type_of_assessment`) had zero variance, containing only a single value across the entire dataset, rendering them uninformative. Regarding the current offense dates, there was not a single row where both `c_offense_date` and `c_arrest_date` were simultaneously non-null, so the two were merged into a single `event_date` column.

Finally, the Asian and Native American racial groups were severely underrepresented, each accounting for less than 1% of the dataset, and were consequently grouped into the existing Other category.

IV. DATA PRE-PROCESSING

A. Column Removal

Beyond the quality issues described above, several additional columns were removed for different reasons.

A first group of columns was deemed irrelevant, as they carry no predictive signal and would only introduce noise: unique identifiers (`id`), free-text name fields (`name`, `first`, `last`), and case number strings (`c_case_number`, `r_case_number`, `vr_case_number`).

A second group consisted of columns redundant with others already retained. Correlation analysis confirmed that `age_cat` is fully captured by `age` and that `score_text` and `v_score_text` are essentially ordinal encodings of

`decile_score` and `v_decile_score`. These were therefore dropped in favour of their numerical counterparts.

Finally, a third group of columns was removed to prevent data leakage. These are features that were only observed *after* the COMPAS score was issued (such as recidivism outcomes, re-arrest dates, and re-incarceration periods) and therefore could not have been available at prediction time. Including them would artificially inflate model performance and invalidate any conclusions drawn.

Additionally, rows where the event date fell more than 30 days before or after the COMPAS screening date were excluded, as such cases likely reflect data entry errors or administrative anomalies rather than genuine records.

B. Feature Engineering

Several new features were derived from the existing variables to enrich the dataset.

- `age_at_event`: the subject's age at the time of the offense or arrest, computed from the date of birth and `event_date`.
- `priors_per_adult_year`: the ratio of prior offenses to the number of adult years, capturing criminal activity rate rather than raw count.
- `total_juv_count`: the sum of all juvenile charges (felony, misdemeanor, and other).
- `has_juv_record`: a binary flag indicating whether any juvenile record exists.
- `first_offense`: a binary flag for subjects with no prior criminal record of any kind.
- `charge_category`: a normalized categorization of `c_charge_desc` into five groups: *violent*, *drug*, *property*, *driving*, and *other*.
- `young_man`: a binary flag for male subjects under 25 years of age.
- `velocity_repeat_offender`: a binary flag identifying subjects with more than one offense per adult year whose current charge is a violent felony.

Categorical variables were then one-hot encoded. For `race`, the `race_other` dummy was dropped as the reference category. For the remaining categoricals, the first dummy was dropped to avoid multicollinearity.

C. Final Dataset

After all pre-processing steps, the dataset was reduced to 6,172 records with no missing values. All features are either numerical or binary-encoded, and the target variable is `decile_score`, a continuous integer score ranging from 1 to 10 representing the COMPAS-assessed recidivism risk.

The final feature set consists of 21 variables. This is notably limited in comparison to the 137 features used by the original COMPAS algorithm, and it is therefore expected that our implementation will fall short of its predictive performance due to the significantly reduced information available to the model.

V. MACHINE LEARNING MODELS

A. Modeling Approach and Setup

Predicting the COMPAS algorithm's `decile_score` (an integer ranging from 1 to 10) presents a unique modeling challenge. It can be framed as a continuous regression problem, a multi-class classification problem, or an ordinal regression problem. To broadly capture the underlying logic, we approached the task primarily from a regression standpoint while testing an ordinal approach.

The dataset was divided into a training set (80%) and a testing set (20%). To ensure that both sets maintained the same distribution of the target variable, a stratified split was performed based on the `decile_score`.

To evaluate the predictive performance of different algorithms, a combination of linear and non-linear models was trained. Hyperparameter optimization was conducted for all models using a 5-fold Cross-Validation approach via Grid Search (`GridSearchCV`).

To evaluate exactly how race attributes influenced the predictions, this entire modeling pipeline was executed twice: once preserving the one-hot encoded race columns (*race-aware*), and once completely dropping them (*race-blind*).

B. Linear Models

Given the structured, tabular nature of the dataset, we initially explored linear relationships, applying scaling transformations and polynomial expansions to capture potential complexities.

a) *ElasticNet with Polynomial Features*: We established a robust linear baseline using ElasticNet, a regularized regression method that linearly combines both L1 and L2 penalties. The model pipeline consists of expanding the dataset to second-degree polynomials, applying standard scaling to normalize the features (to which ElasticNet is highly sensitive), and finally fitting the regressor. Grid search was used to balance the L1 ratio and the regularization strength (*alpha*).

b) *Quantile Regression*: Standard regression models optimize for the mathematical average, which can be sensitive to outliers. Because our dataset exhibits some skewness and focuses on thresholds of high and low risk, we tested Quantile Regression focusing on the median (quantile = 0.5). To further combat target skewness, this base pipeline was wrapped in a `TransformedTargetRegressor`, calculating the natural logarithm of the target before fitting and applying an exponential inverse transformation to the predictions. This stabilizes variance and reduces right skewness.

c) *Ordinal Regression*: Because the `decile_score` is an ordered, discrete scale, we implemented Ordinal Regression using an All-Threshold (AT) Logistic model. This algorithm inherently understands that a prediction of 5 is "greater" than 4, and "less" than 6, avoiding the strict interval distances assumed by standard continuous regression models.

d) *ElasticNet with Splines and Target Transformation*: During exploratory modeling, we observed that linear models seemed unable to break an R^2 score barrier of around 0.5,

suggesting non-linear structures within human behavior characteristics. To give the linear model maximum flexibility, we replaced polynomial features with a `SplineTransformer` (using cubic splines with 5 to 8 knots), coupling it once again with the `TransformedTargetRegressor` (log/exp mappings) and ElasticNet regularization.

C. Non-Linear Models

a) *Random Forest Regressor*: To capture complex, non-linear interactions without relying on explicit feature engineering like polynomials and splines, we trained a Random Forest Regressor. This ensemble method uses a multitude of decision trees and combats overfitting by using a random subset of features (square root of the total features) at each split. The grid search focused on limiting the `max_depth` of the trees, tuning the `min_samples_leaf`, and optimizing the number of estimators (`n_estimators`), ensuring that the model generalizes better to unseen data while capturing nonlinear dependencies inherent to demographic traits and criminal histories.

VI. RESULTS

A. COMPAS Prediction

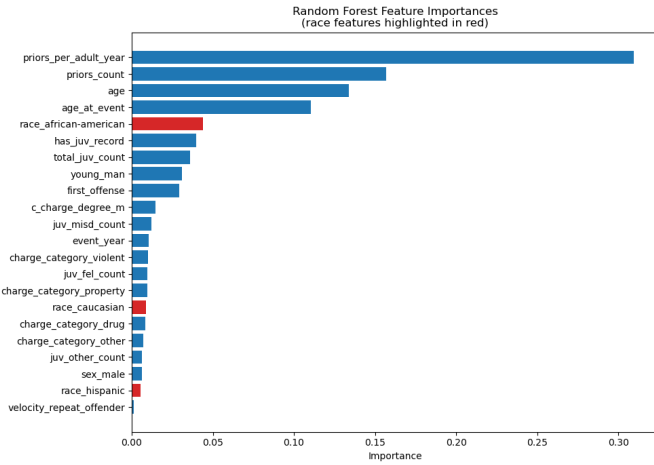


Fig. 3. Random Forest Feature Importance (race-aware model).

Feature	Rank	Importance
race_african-american	4 th	0.0539
race_caucasian	11 th	0.0114
race_hispanic	20 th	0.0052

TABLE I

Race feature importance rankings (race-aware model).

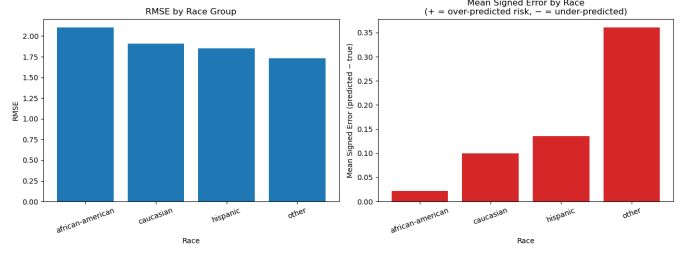


Fig. 4. RMSE and Mean Signed Error Race (race-aware model).

Race	RMSE	Mean Signed Error
African-American	2.1211	0.0266
Caucasian	1.9091	0.0840
Hispanic	1.8889	0.0941
Other	1.7146	0.2523

TABLE II

RMSE and mean signed error by race group (race-aware model).

1) *Race Aware Results*: As we can see in Figure 3 and Table I, being African-American is a roughly 5 times more important deciding factor to determine the COMPAS score than being Caucasian. It is even somehow more important than the existence of a juvenile record or the degree of the charge.

We can also see in on Figure 4 that the RMSE is the highest for the African-American race group. Although not by a huge difference, it shows the algorithm has the most difficulty in classifying this group.

In the same figure, we can see that the model overpredicts for every group. However, considering that African-American already have higher scores in the dataset, the fact that is still overpredicting regardless raises questions.

It is also worth noting that all models, regardless of family or complexity, plateaued at an $R^2 \approx 0.5$. This ceiling is consistent across both race-aware and race-blind conditions.

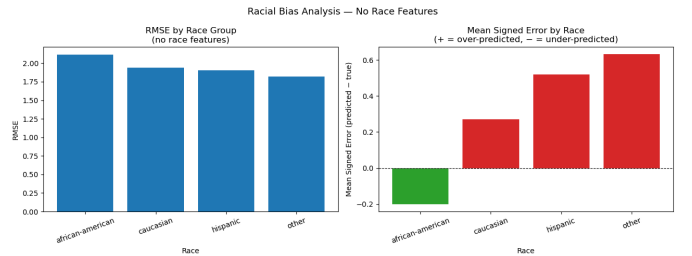


Fig. 5. RMSE and Mean Signed Error by Race (race-blind model).

2) *Race Blind Results*: We can't evaluate the importance of race directly, since this time the model was trained without access to that information.

In Figure 5 we can see that the RMSE stayed roughly the same as in the race-aware model.

However, we can see in the same figure that the model is now actually underpredicting the score for the African-

Race	RMSE	Mean Signed Error
African-American	2.1160	-0.1998
Caucasian	1.9380	0.2706
Hispanic	1.9072	0.5202
Other	1.8165	0.6315

TABLE III

RMSE and mean signed error by race group (race-blind model).

American group. It is also overpredicting significantly more for the other race groups.

B. Recidivism Prediction

To investigate whether the observed disparities originate from the data itself or from the COMPAS scoring methodology, a Random Forest Classifier was trained to predict actual two-year recidivism outcomes (*is_recid*) using the same feature set, again under both race-aware and race-blind conditions.

The classifier achieved a test accuracy of 0.68 and an F1 score of 0.64 under the race-aware condition, with virtually identical results in the race-blind condition.

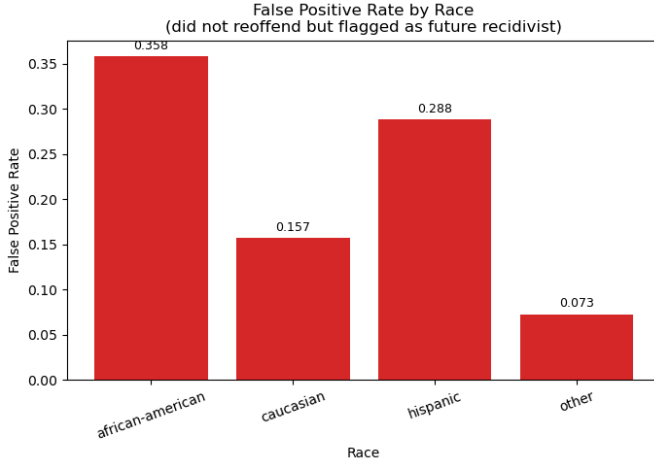


Fig. 6. FPR among race groups (race-aware model).

1) *Race Aware Results:* In Figure 6 we can see the false positive rate for African-Americans is more than two times bigger than for Caucasians.

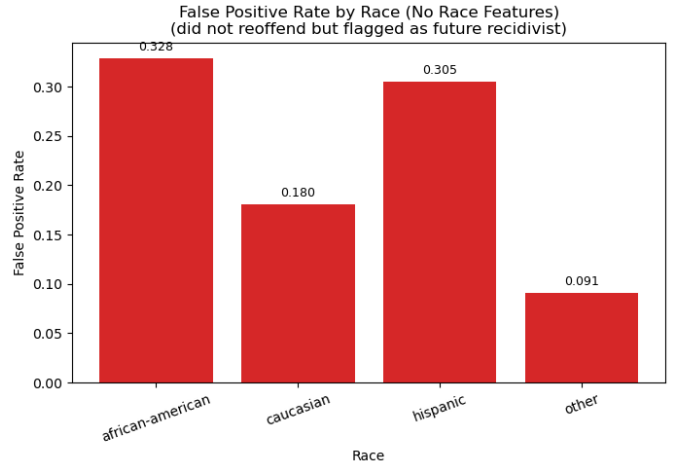


Fig. 7. FPR among race groups (race-blind model).

2) *Race Blind Results:* In Figure 7 we can see the false positive rates for the African-American and Hispanic groups reduced slightly and the rate for Caucasian people increased, but the overall picture is still the same.

VII. CONCLUSIONS

We were able to draw a lot of conclusions from this study regarding the way race influences the model's behavior and also about how the racial disparities persist beyond COMPAS.

A. Race features have negligible impact

All models plateaued at $R^2 \approx 0.5$ regardless of model family, complexity, or whether race features were included, indicating a hard ceiling likely imposed by the limited feature set (21 vs 137 features) rather than modeling choices.

Race features contribute negligible predictive value. Overall performance is virtually unchanged between race-aware and race-blind conditions in both the regression and classification tasks.

Despite contributing little predictive value, *race_african-american* ranked 4th out of 21 features in the race-aware Random Forest (Figure 3), approximately 5 times more important than *race_caucasian*. This indicates race has a strong directional influence on the COMPAS score independently of its predictive utility.

B. Race influences prediction errors across race groups differently

The race-aware COMPAS model overpredicts for all racial groups, but the signed error is lowest for African-Americans despite their scores being the highest in the dataset.

Removing race features causes African-Americans to flip from slightly overpredicted to underpredicted, while overprediction for all other groups increases substantially. Race was therefore suppressing scores for non-Black defendants and inflating them for Black defendants.

C. FPR disparities remain outside of COMPAS

The recidivism classifier produces a false positive rate for African-Americans more than twice that of Caucasians, even when trained on actual outcomes rather than COMPAS scores.

This FPR disparity persists when race features are removed, indicating the bias is not introduced solely by explicit race information but is embedded in correlated features within the data.

The persistence of FPR disparity in the outcome-based classifier suggests the bias is at least partially structural, present in the data independently of COMPAS, rather than being exclusively a product of the algorithm's scoring methodology.

While structural bias in the data accounts for a significant portion of the observed disparities, the disproportionate importance of `race_african-american` in predicting the COMPAS score suggests the algorithm introduces additional bias beyond what the data alone would produce.

VIII. CRITICAL REFLECTION

A. Racial Disparities Beyond COMPAS

The false positive rate disparity observed in our recidivism classifier, trained on actual two-year outcomes rather than COMPAS scores, demonstrates that racial disparities in risk prediction are not unique to the COMPAS algorithm. A model with no knowledge of COMPAS reproduces similar patterns of disproportionate misclassification, suggesting the disparities are rooted in the data itself rather than the scoring methodology.

This is consistent with Han et al. [5], who apply a causal framework to the same dataset and find that racial disparities in recidivism outcomes persist beyond what algorithmic scores alone can explain, attributing the remainder to structural inequalities in housing, employment, and social support that disproportionately affect minority groups over time.

B. Proxy Variables and Structural Bias in the Data

Even without explicit race features, our models continue to produce racially disparate error patterns. This is consistent with the concept of feature bias described by Neil and Zanger-Tishler [6]: because prior arrest records are among the most predictive features in risk assessment instruments, and because racial differences in the likelihood of arrest given the same underlying offending behavior are well documented, the racial signal is already embedded in these features before any model is trained.

This is directly observable in the composition of our dataset. The Broward County Prosecutorial Performance Indicators report [7] documents that Black defendants are the most likely to receive custodial sentences and the least likely to have felony charges reduced after filing (prosecutorial disparities that directly determine who accumulates a criminal record, and therefore who enters the dataset and with what criminal history).

C. Algorithmic Amplification of Structural Bias

The deployment of risk assessment instruments creates a feedback loop that extends beyond any individual score. Ensign et al. [8] formalise this dynamic: predictions derived from historically biased arrest data direct enforcement toward the same communities, generating more arrests that feed back into future assessments. Our results support this concern. A classifier trained on actual recidivism outcomes still produces substantial FPR disparities across racial groups, meaning the bias persists independently of COMPAS's scoring methodology. Auditing or replacing the algorithm alone would not resolve it.

AI TOOLS ACKNOWLEDGMENT

Throughout the development of this project, we used AI tools, namely GitHub Copilot (powered by Gemini and Claude), to assist us with data exploration, model selection and document formatting.

Specifically, we used these tools to brainstorm modeling approaches. They also saved us significant time by helping automate repetitive pipelines, setting up the grid search, and translating our findings into the proper LaTeX structure (when needed).

However, we want to emphasize that these tools only provided technical support, and the research and critical thinking was provided by us.

REFERENCES

- [1] J. Angwin, J. Larson, S. Mattu, and L. Kirchner, "Machine Bias: There's software used across the country to predict future criminals. And it's biased against blacks," *ProPublica*, May 2016. Available: <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>.
- [2] W. Dieterich, C. Mendoza, and T. Brennan, "COMPAS Risk Scales: Demonstrating Accuracy Equity and Predictive Parity," *Northpointe Inc.*, Jul. 2016. Available: https://go.volarisgroup.com/rs/430-MBX-989/images/ProPublica_Commentary_Final_070616.pdf.
- [3] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent Trade-Offs in the Fair Determination of Risk Scores," *arXiv preprint arXiv:1609.05807*, 2016.
- [4] A. Chouldechova, "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments," *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.
- [5] J. X. Han, K. Greenewald, and D. Shah, "Fairness Is More Than Algorithms: Racial Disparities in Time-to-Recidivism," *arXiv preprint arXiv:2504.18629*, 2025.
- [6] R. Neil and M. Zanger-Tishler, "Algorithmic Bias in Criminal Risk Assessment: The Consequences of Racial Differences in Arrest as a Measure of Crime," *Annual Review of Criminology*, vol. 8, pp. 97–119, 2025. <https://doi.org/10.1146/annurev-criminol-022422-125019>.
- [7] Prosecutorial Performance Indicators, "Race and Prosecution in Broward County, Florida," Florida International University and Loyola University Chicago, Nov. 2022. Available: <https://prosecutorialperformanceindicators.org/wp-content/uploads/2022/11/PPI-Race-and-Prosecution-in-Broward-County-Florida-Report.pdf>.
- [8] D. Ensign, S. A. Friedler, S. Neville, C. Scheidegger, and S. Venkatasubramanian, "Runaway Feedback Loops in Predictive Policing," in *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, vol. 81, pp. 160–171, 2018. Available: <https://proceedings.mlr.press/v81/ensign18a.html>.